PaisaPro.ai

AI-Powered Investment Advisory Platform

for the Indian Equity Market

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Agenda

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3 System Architecture

4 Analytics Engine

5 ML & Forecasting

6 Agentic AI Advisor

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The Problem: Why We Built PaisaPro.ai

150M+
Demat accounts in India (2024)

INR 20K Cr
Monthly SIP inflows

Yet retail investors lack institutional-grade analytical tools.

Three Critical Gaps

1. **Analytics Gap**
   ARIMA, GARCH, Fama-French require deep expertise
2. **AI Advisory Gap**
   LLMs lack live data & portfolio context
3. **Tool Fragmentation**
   Screener + TradingView + Groww + ChatGPT

No single platform unifies all of these

Our Solution: Four Pillars

Quantitative Finance

ARIMA · GARCH
Fama-French 4-Factor
Markowitz MVO

Monte Carlo (1K sims)

Machine Learning

Random Forest Classifier
3-Model Regression
Isolation Forest

K-Means Clustering

Agentic AI Advisor

Llama 3.3 70B (Groq)
4 Live Tools
Portfolio Context

SSE Streaming

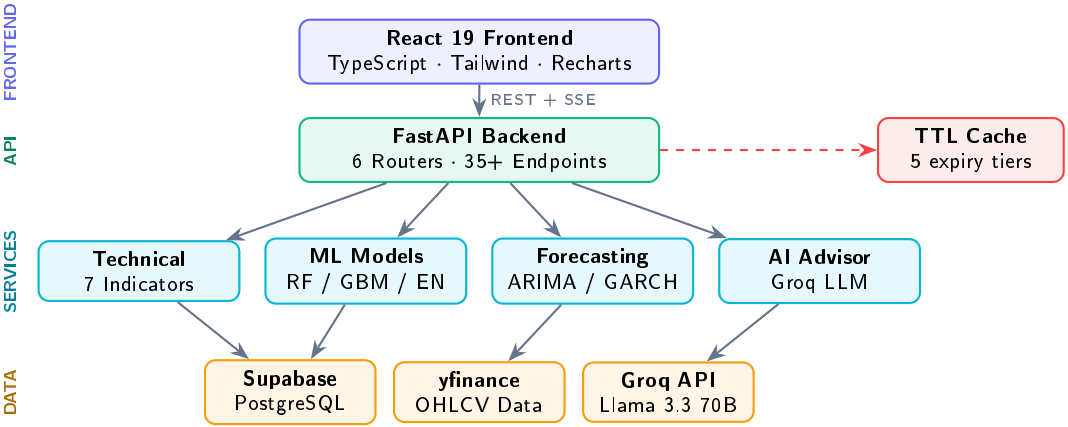
Full-Stack Platform

React 19 + FastAPI
18 Pages · 35+ APIs
Dark/Light Theme

Cloud Deployed

500+ NSE Stocks · Daily Auto-Refresh · Sub-100ms Cached Responses

System Architecture



19 service modules · Docker on HF Spaces · Netlify CDN · Supabase RLS · Daily cron pipeline

Data Pipeline & Database Design

Daily Refresh Pipeline

1. APScheduler triggers at **6 PM IST**
2. Clear all in-memory caches
3. Batch-fetch OHLCV via yfinance
4. Upsert into Supabase (500+ stocks)
5. Compute 30+ technical indicators
6. Refresh macro data (VIX, INR, Gold, Oil)
7. Pre-warm analytics cache
8. Retrain 500+ RF classifiers

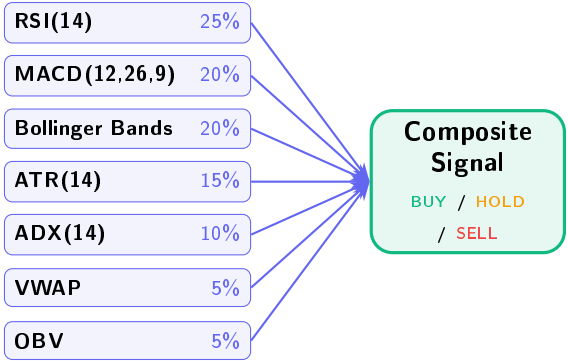
Database (Supabase PostgreSQL)

Table	PK	Purpose
stocks	symbol	Universe metadata
stock_prices	(symbol, date)	Daily OHLCV + indic
macro_prices	(ticker, date)	VIX, INR, Gold, Oil
portfolio_holdings	id	User holdings
watchlist	id	User watchlist

Multi-Tier Cache (5 TTL levels)

Indices	5 min
News / Macro	30 min
Volatility / Sector	1 hour
Analytics report	2 hours
Price DataFrames / ML	24 hours

Analytics Engine: 7-Indicator Composite Signals



Signal Classification

$$Score = \sum_{i=1}^{7} w_i \cdot s_i$$

- $Score > +0.3 \rightarrow$ **BUY**
- $Score \in [-0.3, +0.3] \rightarrow$ **HOLD**
- $Score < -0.3 \rightarrow$ **SELL**

Typical Distribution (Nifty 500)

BUY	120–180 stocks	24–36%
HOLD	200–280 stocks	40–56%
SELL	80–140 stocks	16–28%

Machine Learning Pipeline

Random Forest Classifier

Task: 5-day directional prediction

Trees	100
Training window	252 days (1 year)
Features	RSI, MACD, BB, vol ratio, etc.
Accuracy	**52–58%** (above 50% baseline)

Unsupervised Models

K-Means (k=4):

Defensive · Value · Momentum · High-Volatility

Isolation Forest:

Detects price spikes, volume anomalies, earnings events

3-Model Regression Ensemble

Task: 30-day return forecasting

Random Forest	200 trees, depth 6
ElasticNet	L1+L2, RobustScaler
HistGBM	100 est., LR 0.05

Key innovations:

- 32+ features (OHLC microstructure)
- Vol-adjusted log-return targets
- Walk-forward CV with embargo gap
- Sharpe-weighted ensemble

Directional accuracy: **55–60%**

Time-Series & Volatility Forecasting

ARIMA Forecasting

- **Stationarity:** ADF + KPSS dual tests
- **Grid search:** $p \in \{0..3\}$, $d \in \{0..2\}$, $q \in \{0..3\}$
- **Selection:** AIC minimization (48 candidates)
- **Output:** 30-day forecast + 95% CI

Holt-Winters ETS

$$\hat{y}_{t+h} = l_t + \sum_{j=1}^{h} \phi^j \cdot b_t$$

- Damped additive trend

GARCH(1,1) Volatility Model

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

- Captures **volatility clustering**
- Persistence $\alpha + \beta \approx 0.95$
- 30-day conditional volatility forecast
- MLE via `arch` library

Volatility Regime Signals

LOW	$<$ 25th pctl	$\rightarrow$ Entry signal
NORMAL	25th–75th	$\rightarrow$ Neutral
HIGH	75th–95th	$\rightarrow$ Caution
EXTREME	$>$ 95th	$\rightarrow$ Caution

Risk Analytics & Portfolio Optimization

Fama-French 4-Factor Model

$$R_i - R_f = \alpha + \beta_1 MKT + \beta_2 SMB + \beta_3 HML + \beta_4 MOM$$

Market	Nifty 50 excess returns
SMB	Small − Big (size)
HML	High − Low (value)
MOM	Winners − Losers

Output: α, betas, t-stats, R^2

Markowitz MVO

$$\min_{\mathbf{w}} \ \mathbf{w}^T \Sigma \mathbf{w} \quad \text{s.t.} \ \sum w_i = 1, \ w_i \leq 0.4$$

- 2,000 Monte Carlo random portfolios
- SLSQP constrained optimization
- Min-variance & max-Sharpe identification
- Efficient frontier scatter visualization

Risk Metrics Suite

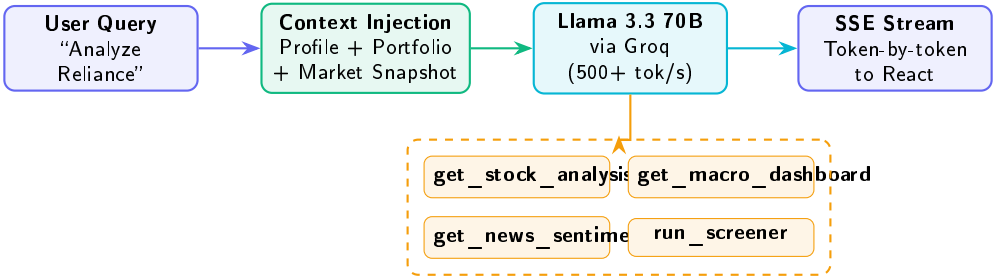
Sharpe · Sortino · Beta · Alpha · VaR(95%) · Max Drawdown · Annualized Volatility

Risk-free rate: **6.5%** (10Y G-Sec yield)

Monte Carlo Wealth Planner

- 1,000 simulations per scenario
- SIP with annual step-up
- P10/P25/P50/P75/P90 bands

Agentic AI Advisor — The Key Differentiator



4 Live Function-Calling Tools (up to 3 autonomous rounds)

What Makes It "Agentic"?

- LLM **autonomously decides** when to call tools
- Multi-turn reasoning with tool results

Beyond Static RAG

- Static RAG: inject data → generate
- **Agentic**: reason → decide tool → fetch data → synthesize → respond

Technology Stack

Backend

Python	3.12
FastAPI	0.111
pandas	2.2
scikit-learn	1.8
statsmodels	0.14
arch	8.0
scipy	1.17
yfinance	0.2
Pydantic	v2

Frontend

React	19
TypeScript	5.9
Vite	8.0
Tailwind CSS	4
Recharts	3.8
Zustand	5.0
React Query	5
React Router	7
Axios	1.13

Infrastructure

Database	Supabase
Backend	HF Spaces
Frontend	Netlify CDN
LLM	Groq Cloud
Container	Docker
Model	Llama 3.3 70B
Scheduler	APScheduler

Scale

19 service modules
6 API routers
35+ endpoints
18 frontend pages
5-tier TTL cache

18 Interactive Pages

Core

- Dashboard
- AI Advisor Chat
- Portfolio Manager
- Analytics Report
- Stock Screener
- Portfolio Optimizer

Advanced Analytics

- Time-Series Forecast
- Volatility Forecast
- Sector Rotation
- Macro Dashboard
- Risk Factors
- ML Prediction

Planning & Extras

- Forward Planner
- Goal Planner
- Scenario Compare
- News Sentiment
- Model Health (MLOps)
- Landing Page

✓ Dark/Light themes ✓ Responsive (320px – 2560px) (Recharts) ✓ 7 chart types

✓ SSE streaming chat ✓ Markdown rendering ✓ WCAG 2.1 AA

Results & Validation

System Performance

Stocks analyzed daily	**500+**
API endpoints	**35+**
Frontend pages	**18**
Cached response latency	**< 100ms**
Cold analytics build	**30–60s**
LLM first token	**2–5s**

Testing Results

API endpoint tests	**35/35 passed**
Edge case tests	**15/15 passed**
Frontend page tests	**18/18 passed**
Cross-browser (6 browsers)	**All pass**

Model Performance

RF Classifier accuracy	**52–58%**
Regression dir. accuracy	**55–60%**
GARCH persistence	**∼0.95**
ARIMA best-model rate	**∼60%**

Data Integrity

Metric	vs.	Dev
RSI(14)	TradingView	<0.5%
MACD	TradingView	<0.3%
BB	TradingView	<0.1%
Prices	Yahoo Finance	0%

Comparison with Existing Platforms

Capability	PaisaPro	Screener.in	TradingView	Groww	ChatGPT
Technical Signals (7 indicators)	✓	✗	✓	✗	✗
ML Classification	✓	✗	✗	✗	✗
ML Regression Ensemble	✓	✗	✗	✗	✗
ARIMA / ETS Forecast	✓	✗	✗	✗	✗
GARCH Volatility	✓	✗	✗	✗	✗
Fama-French 4-Factor	✓	✗	✗	✗	✗
Portfolio Optimization	✓	✗	✗	✗	✗
Monte Carlo Planning	✓	✗	✗	✗	✗
AI Advisory (Live Data)	✓	✗	✗	✗	~
Agentic Tool Use	✓	✗	✗	✗	✗
Portfolio Tracking	✓	✗	✗	✓	✗
News Sentiment	✓	✗	✗	✗	✗

Limitations & Future Work

Current Limitations

1. Single-user mode (hardcoded ID)
2. Daily OHLCV only (no intraday)
3. Keyword-based sentiment (not FinBERT)
4. Proxy Fama-French factors (not academic)
5. Normal distribution in Monte Carlo
6. LLM hallucination risk

Proposed Future Enhancements

1. JWT multi-user authentication
2. FinBERT transformer sentiment
3. 5-min / 15-min intraday data
4. Fama-French 5-factor model
5. Options analytics (Greeks, IV)
6. React Native mobile app
7. Real-time WebSocket feeds
8. Fundamental analysis (P/E, debt ratios)
9. SEBI RIA compliance exploration

Key Lessons Learned

Engineering

- **Caching is critical for UX** — without it, 30–60s per request; with it, <100ms
- **Service-oriented modularity** — 19 independent modules enabled rapid iteration
- **SSE transforms perceived performance** — users start reading within 2s vs waiting 10–15s

Research

- **Model simplicity wins** — RF/GBM outperformed deep learning for daily data (250–1250 obs)
- **Agentic prompt engineering is iterative** — balancing over-eager vs insufficient tool use
- **Financial data quality is variable** — robust NaN handling essential for production

Conclusion

500+
NSE Stocks Analyzed Daily

19
Backend Service Modules

4
Agentic AI Tools

14/14
Objectives Achieved

PaisaPro.ai demonstrates that institutional-grade financial analytics can be democratized for 150M+ Indian retail investors

through the convergence of quantitative finance, machine learning, and agentic AI

Thank You!

Questions & Discussion

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